

An Interactive Dashboard Report on “AI IMPACT IN FUTURE JOB MARKETS BY 2030”

OVERVIEW:

Created an interactive dashboard using **Microsoft Power BI** platform / tool. **The dashboard is prepared for the analysis, comparisons and identifying the trends of AI Impact on Job Market from 2026 – 2030.** Which is titled as **AI Impact in Future Job Markets By 2030.** The **DAX (Data Analysis Expressions)** are used in this project to create **calculated measures and calculated column.** The dataset used here is in **CSV (Comma - Separated Values)** format. Which is sourced from **Kaggle (Public Dataset).** The project consist of **three interactive dashboards / sheets.**

Dashboard / Sheet – 1: **Title Page.**

Dashboard / Sheet – 2: **Overall Overview.**

Dashboard / Sheet – 3: **AI Impact Analysis.**

The Power BI dashboards and reports are published to GitHub for easy access and future enhancement.

GitHub Repository: [Link](#)

INTRODUCTION:

Artificial Intelligence has become one of the most transformative technologies in today's world. Organizations and Industries are adopting to AI - powered tools and automations to **improve efficiency, enhance productivity, and support data – driven and decision – making.** While these advancements create new opportunities, but it also introduce **challenges** such as **workforce transformation, changing job requirements, skill gaps, and concerns regarding job displacement.**

So to understand these technologies, advancements, opportunities and challenges, this project **AI Impact In Future Job Market By 2030** consisting of interactive dashboards which provides trends from **2026 – 2030.** Where user can get **meaningful insights, trends and comparisons across various sectors and fields.** The dashboard consist of visualizations which are easy understand.

The main aim of this project lies on the relationship between AI and employment, by comparing AI Replacement Risk, Future Demand Scores, Automation Trends, AI Tools Usage, Remote Work Adoption, and Workforce Readiness. This project supports for **Business Analytics** and much better understanding of AI and its impact.

TOOLS & TECHNOLOGIES:

- Microsoft Power BI Desktop.
- Dataset Source (Kaggle).
- Power Query Editor.
- Data Analysis Expressions (DAX).
- Color Hunt (Colour Palettes).
- Open Source AI (Reference).

Microsoft Power BI Desktop:

Microsoft Power BI Desktop is the primary tool used to create and develop the **dashboards**. It is used for importing, storing and transforming the data, creating data models, writing the DAX expressions and developing interactive visuals such as (cards, charts, maps and filters etc...).

Dataset Source:

The dataset used in this project is obtained from **Kaggle** contains information related to **workforce characteristics, AI adoption, automation levels, employee performance, salary distribution, future job demand, hiring trends, and AI replacement risk**. The dataset used here is in **CSV (Comma - Separated Values)** format.

Power Query Editor:

While loading the dataset the power query editor is used for data preparation, organize the dataset before analysis and storing the data. **The tasks performed are verifying data types, handling missing values, renaming columns where necessary, and ensuring data consistency to improve the overall quality of the data.**

Data Analysis Expressions (DAX):

DAX is used to create calculated measures and calculated column that supports advanced analysis within the dashboard. Cards having **Average AI Replacement Risk, Average AI Tool Usage, Upskilling Required Percentage, Average Future Demand Score** are developed by using DAX formulas to provide meaningful business insights.

Color Hunt:

Customized themes are used while preparing the dashboards by choosing various colour palettes from colour hunt. The colours are applied to all the visual elements (**including cards, charts, maps, icons and buttons**). The two main colours used for the dashboard background are **#1079B6** and **#6746B1** which is choose from the **project logo**.

DASHBOARDS:

DASHBOARD – 1: TITLE PAGE

The dashboard – 1 consist of project logo along with the title and subtitle included.

The project titled as “**AI Impact In Future Job Maret By 2030**” by using the colour template #1079B6 (dark, vivid cyan – blue colour). And the subtitle as “**Workforce Analytics Dashboard**” by using the colour template #6746B1 (deep iris purple).

Also a **button (view dashboard) at the right bottom is created** for navigating to the next dashboard / sheet (Overall Overview). **Which can be accessed by clicking the button** or by pressing **cmd + click** (for mac) and **control + click** (for windows).

“The dashboard / sheet – 1 is designed clearly to highlight the project logo and title, followed by subtitle.”

DASHBOARD – 2: OVERALL OVERVIEW

The Overall Overview dashboard consist of 13 visuals in total. Each visual is given a specific colour palette including with the primary colour of the page. The dashboard has **(five – Cards, a Dropdown Slicer, List Slicer, 100% Stacked Column Chart, Area Chart, Map Chart, Pie Chart, Line Chart and Tree Map).**

CHART COMPARISONS:

1. **Cards:** Total Employees, Average Salary, Average Performance, Average Job Satisfaction and Average Experience.

Total Employees – 3000

Average Salary – \$127K

Average Performance – 3.51

Average Job Satisfaction – 3.0

Average Experience – 1.49K

“The cards also include images representing the title of the card.”

2. **Dropdown Slicer:** Includes **JOB TITLES** of the dataset. They are total 20 job titles present.
3. **List Slicer:** Providing the **LIST OF COUNTRIES**. They are total 10 countries present in dataset.
4. **100% Stacked Column Chart:** Provides the comparison between **HIRING TREND** vs **AUTOMATION LEVEL** by **WORKFORCE COUNT**.
5. **Area Chart:** Provides the comparison between **JOB SATISFACTION** by **COMPANY SIZE** from **WORKFORCE COUNT**.

6. **Map Chart:** Provides **WORKFORCE SALARY** by **COUNTRY WISE**.
7. **Pie Chart:** Gives the relation between **WORKFORCE DISTRIBUTION** by **EDUCATION LEVEL**.
8. **Line Chart:** Provides the comparison between **AVERAGE SALARY (\$)** by **YEARS of EXPERIENCE** along with **AVERAGE JOB SATISFACTION**.
9. **Tree Map:** Shows **EMPLOYEES COUNT** by **DIFFERENT INDUSTRIES**.

DAX FORMULAS USED:

1. **TOTAL EMPLOYEES:**
Total Employees = COUNT('table_name', Employee_ID)
2. **AVERAGE SALARY:**
Average Salary = AVERAGE('table_name'[Average_Salary_USD])
3. **AVERAGE PERFORMANCE:**
Average Performance = AVERAGE('table_name'[Performance_Score])
4. **AVERAGE JOB SATISFACTION:**
Average Job Satisfaction = AVERAGE('table_name'[Job_Satisfaction])
5. **AVERAGE EXPERIENCE:**
Average Years Experience =
CALCULATE(COUNTROWS('table_name'),'table_name'[Upskilling_Needed] = "Yes")

DASHBOARD – 3: AI IMPACT ANALYSIS

The AI Impact Analysis dashboard consist of **19 visuals in total**. Each visual is given a specific colour palette including with the primary colour of the page. The dashboard has (**five – Cards, seven – Dropdown Slicers, 100% Stacked Column Chart, Map Chart, Clustered Column Chart, two – Donut Charts, Stacked Bar Chart and Tree Map**).

CHART COMPARISONS:

1. **Cards:** Average AI Risk, Average Future Demand, High AI Tools Usage, Upskill Required, Average Job Growth.
Average AI Risk – 0.50
Average Future Demand – 0.60
High AI Tools Usage – 1.014K
Upskill Required – 49.63%
Average Job Growth – 17.24

“The cards also include images representing the title of the card.”

2. **Dropdown Slicer:** There are seven columns chosen for each slicer.
(Industry, Job Title, Country, Required Skills, Education level, AI Tools Usage, Years of Experience).

3. **100% Stacked Column Chart:** Provides the comparison between **HIRING TREND** by **COMPANY SIZE** with **TOTAL EMPLOYEES**.
4. **Map Chart:** Provides **FUTURE DEMAND SCORE** by **COUNTRY**.
5. **Clustered Column Chart:** Gives the comparison between **AVERAGE PERFORMANCE** by **AI TOOLS USAGE**.
6. **Donut Charts:** **First Donut Chart** gives the comparison between **FUTURE READY STATUS** by **TOTAL EMPLOYEES**. And the **Second Donut Chart** give the comparison between **AI TOOLS USAGE** by **TOTAL EMPLOYEES**.
7. **Stacked Bar Chart:** Gives the comparison for **COUNTRIES** by **REMOTE WORK ADOPTION** with **TOTAL EMPLOYEES** and also including ribbon visualization to the chart.
8. **Tree Map:** Provides the comparison for **AVERAGE AI REPLACEMENT RISK** by **INDUSTRIES**.

DAX FORMULAS USED:

1. **AVERAGE AI REPLACEMENT:**
Average AI Replacement Risk = AVERAGE('table_name'[AI_Replacement_Risk])
2. **AVERAGE FUTURE DEMAND:**
Average Future Demand Score = AVERAGE('table_name'[Future_Demand_Score])
3. **HIGH LEVEL AI TOOLS USAGE:**
High Level AI Tools Usage =
COUNTROWS(FILTER('table_name','table_name'[AI_Tool_Usage] = "High"))
4. **UPSKILLING REQUIRED:**
Upskilling Required =
DIVIDE(CALCULATE(COUNTROWS('table_name'),'table_name'[Upskilling_Needed] = "Yes"),COUNTROWS('table_name'))
5. **AVERAGE JOB GROWTH:**
Average Job Growth 2030 = AVERAGE('table_name'[Job_Growth_2030])

DAX FORMULA FOR CALCULATED COLUMN:

1. **FUTURE READY STATUS:**
Future Ready Status = IF('table_name'[AI_Tool_Usage] = "High" &&
'table_name'[Upskilling_Needed] = "No", "Future Ready", "Needs Improvement")
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CONCLUSION:

The project “**AI Impact in Future Job Market by 2030**” is successfully completed. And it demonstrates how business intelligence can be able to analyse the workflow data and provide meaningful insights through interactive visualization in dashboards by using **Microsoft Power BI**. The raw dataset are loaded into power query and transformed. The project consist of two main dashboards **Overall Overview providing the summary of the project** and **AI Impact Analysis focusing on different aspects / fields in AI**. Throughout the project **DAX** calculations were applied for (**calculated measure and calculated column**). Along with specific **colour palettes** chosen for each visualization and dashboards.

Overall, this project highlights the value of data visualization by turning raw dataset into meaningful insights. Which supports in real world decision – making and understanding complex data.

FUTURE SCOPE:

This project can be further enhanced by integrating more real – time data sources to provide up – to – date workforce insights. Additional features such as **advanced Power BI functionalities can be implemented to improve accuracy and analysis.** Also dashboard features such as **drill – through pages, custom tooltips, bookmarks, and advanced navigation** can improve user experience and enable more detailed exploration of insights. **Overall this project has a strong potential for future development and can be able to provide more data – driven solutions.**
